

CLiC-it2026 · Palermo, Italy · September 14–16, 2026

Lost in Transcription

Investigating the Role of Audio Information for Video-based Visual Statement Verification



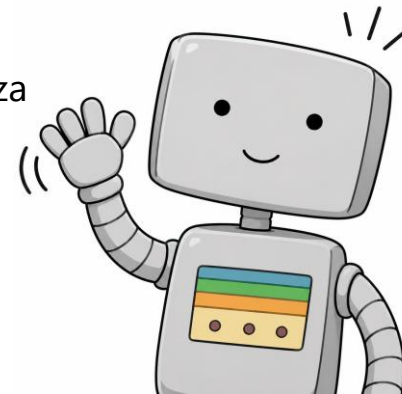
Michela Deodati^{1,†} · Mohd Kaif^{2,†} · Davide Testa^{3,4} · Giovanni Bonetta³ · Alessandro Lenci¹ · Oswald Lanz² ·
Raffaella Bernardi² · Bernardo Magnini³

¹University of Pisa · ²Free University of Bozen-Bolzano · ³Fondazione Bruno Kessler (FBK) · ⁴University of Rome La Sapienza

[†]These authors contributed equally



SAPIENZA
UNIVERSITÀ DI ROMA



Motivation

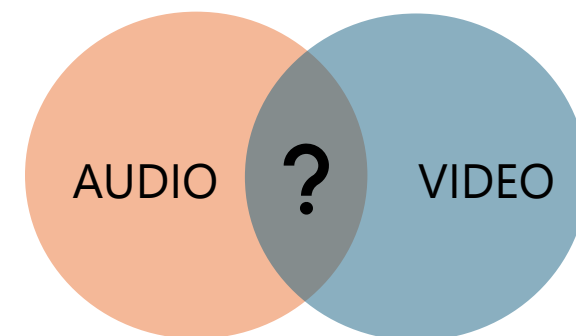
- Omnimodal LLMs have evolved to **process vision, language, and raw audio collectively**.
- Acoustic cues **enhance the visual experience**.
- Recent benchmarks have not isolate audio signal and **examine the specific causal role** that audio plays in forming the answer. ([Fu et al., 2025](#); [Geng et al., 2025](#))
- **Integration** is not **certain**

?

How do the two modalities interact during reasoning?

>

Does audio truly aid multimodal video reasoning?



Research Question



RQ-1:

To what extent does raw audio affect video-based VSV when jointly processed with visual and textual information?

RQ-2:

How effectively do multimodal models integrate visual, textual and auditory information for video-based reasoning?

CONTRIBUTIONS

- 1 Extend MAIA with an auxiliary audio layer: transcribed content plus a coarse acoustic-context label for every video
- 2 Examine audio from two perspectives: as a raw signal using an omnimodal model, and as text through automatic transcription.
- 3 Run controlled experiments and targeted ablations to isolate how visual, textual and auditory signals interact

MAIA & the VSV Task:

- **MAIA**: native-Italian, video-based, competence-oriented benchmark
([Testa et al., 2025](#))
- **MAIA** was built around **visual grounding**, audio was never part of the original benchmark design
- 100 videos · 12 reasoning categories · 2,400 questions, each with a pool of 8 human answers
- **Visual Statement Verification** (VSV): given a video, pick the TRUE statement from a minimally-contrastive **True/False** pair

VSV PROMPT

<video> Choose the correct description:

A. {{ True Statement }}

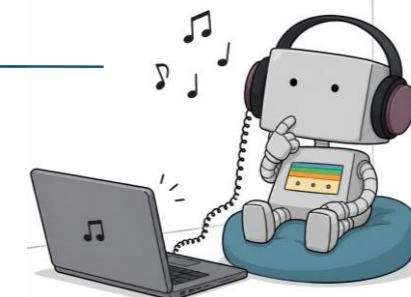
B. {{ False Statement }}

Answer only A or B.

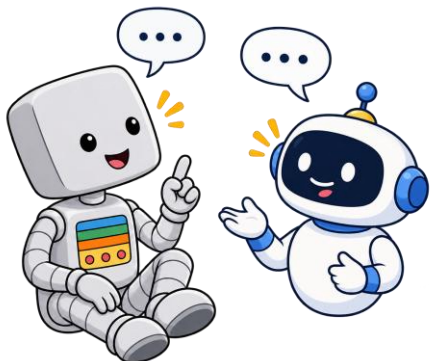
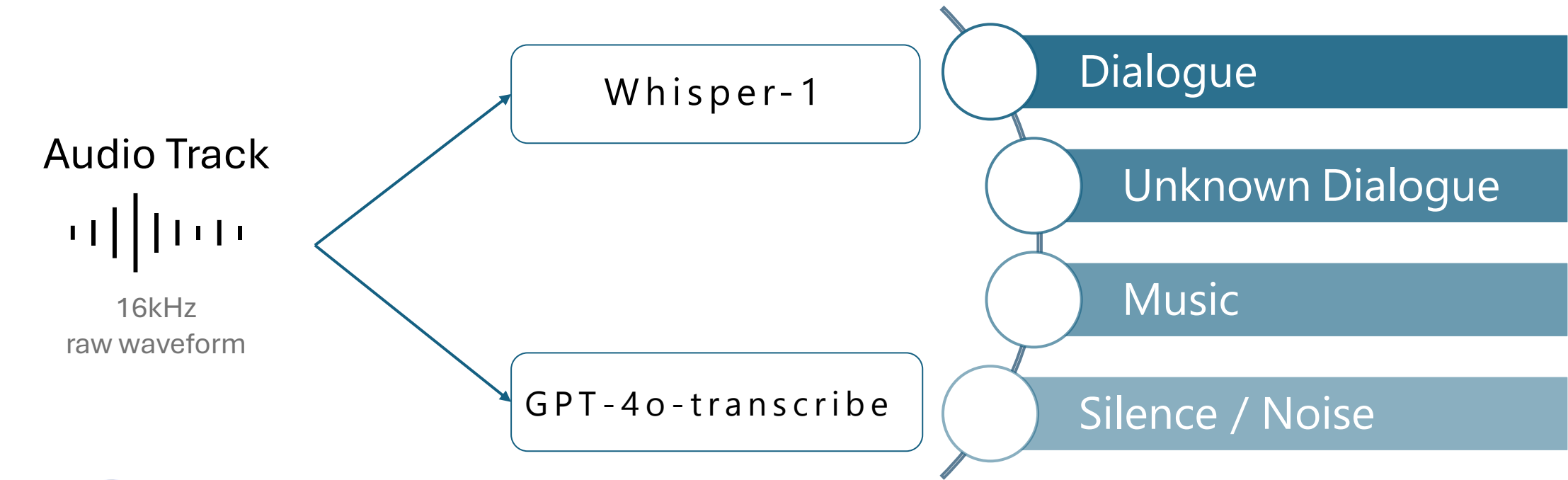
EXAMPLE:

A. Nel video, il pizzaiolo appare concentrato mentre tira fuori la pizza dal forno.

B. Nel video, il pizzaiolo appare distratto mentre tira fuori la pizza dal forno.

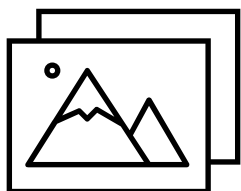


Building an Audio-Transcription Layer:

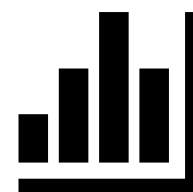


Models & Setup:

Model	Role
Qwen2.5-Omni-7B	Main experiment with raw Audio (omnimodal)
Qwen2.5-VL-7B	Main experiment with Transcriptions (vision-language, no native audio)



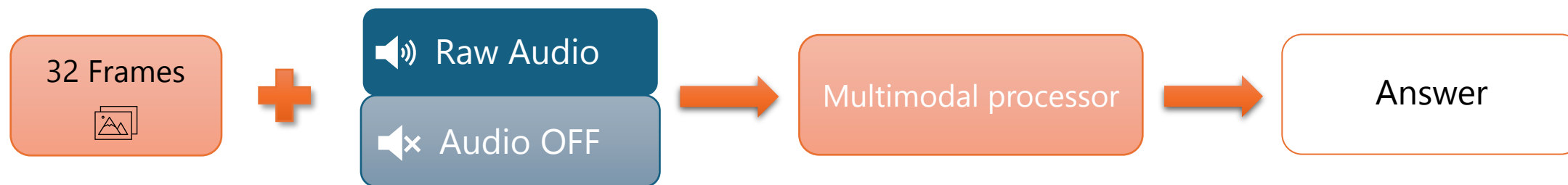
32 frames uniformly
sampled from every
video, across all
experiments



Zero-shot inference · strict
A/B multiple-choice VSV
prompt for every
configuration

Experiment 1: Raw Audio (Qwen2.5-Omni-7B)

*Every video is processed twice, once with audio, once without, so its effect can be **isolated video by video**.*



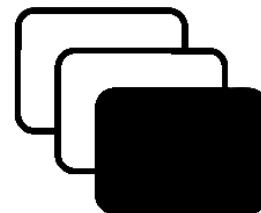
Ablation: Removing Visual Information

*Two visual-density settings, each tested with audio ON and OFF, **isolate how much audio matters as visual evidence shrinks***



Black Frame

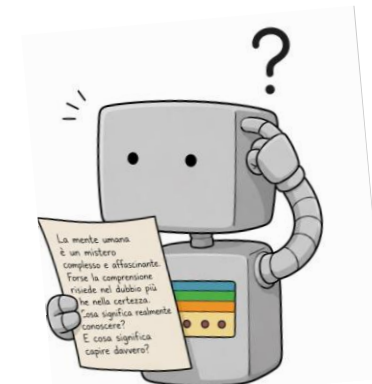
- Isolates pure audio
- No visual grounding is available at all
- Proxy for a “no-video” condition



1-Frame

- A minimal but non-zero visual anchor
- Removes temporal visual motion
- Tests the “static appearance” condition

Experiment 2: Transcriptions (Qwen2.5-VL-7B)



Unimodal Baseline

Linguistic-plausibility bias in the answer options alone

Transcription-only

Whether transcript text supports the decision on its own

32 Frames (w/o Transcription)

Visual grounding performance alone

32 Frames (with Transcription)

Whether the transcript adds anything once video is present

RESULTS

Visual Dominance

RAW AUDIO • QWEN2.5-OMNI-7B

-0.02 pp

82.70% → 82.68%

not significant • $p = .620$

No acoustic category is significant after Holm correction. Raw audio is effectively ignored when visual evidence is dense.

TRANSCRIPTION • QWEN2.5-VL-7B

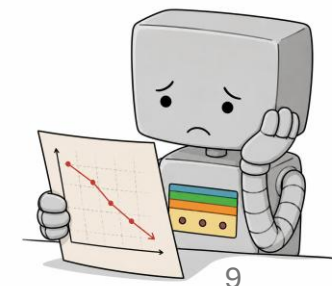
-1.89 pp

80.67% → 78.78%

highly significant • $p = 3.6 \times 10^{-17}$

Significant in every single acoustic category after Holm correction — a systematic effect, not a few outlier items.

Under dense visual evidence, **raw audio is neutral**, but **transcribed audio is not**.



Linguistic Bias & the “Dialogue Paradox”

TRANSCRIPTS CAN BE WORSE THAN NO INFORMATION AT ALL

Unimodal Baseline (no video, no audio) **68.72%**

Transcription-only (transcript, no video) **63.98%**

Reading the transcript alone performs worse than guessing from the question's linguistic plausibility.

-4.74 pp
vs. baseline

EVEN CLEAR, INTELLIGIBLE SPEECH DOESN'T RELIABLY HELP

Dialogue category — raw audio -0.29 pp (n.s.)

Dialogue category — transcription **-1.16 pp (significant, p_Holm = .005)**

Even **understandable speech** can actively mislead **once visual evidence is already dense.**

Qualitative Case Studies

VIDEO 37 • LARGEST IMPROVEMENT

audio **+2.60 pp** (18 Wrong→Correct)

transcript **+1.56 pp** (8 Wrong→Correct)

A child falls and starts crying. Audio links the crying to the fall itself, not to a later, unrelated action — fixing a causal / temporal question the visual stream alone left ambiguous.

VIDEO 71 • LARGEST DROP

transcript **-5.73 pp** (15 Correct→Wrong)

audio **-4.17 pp** (12 Correct→Wrong)

A street vendor sells oranges. Audio is topically coherent (“oranges”, “beautiful and sweet”) but not fine-grained enough — it misleads the model on tool-use and timing questions.

Audio helps only when it is directly **aligned with the exact discriminative detail** the task needs.

RESULTS

The Visual Threshold

BLACK FRAME • NO VISUAL INFO

+5.79 pp

67.40% → 73.19%

highly significant • $p = 2.44 \times 10^{-76}$

Robust in ALL 4 categories: Dialogue +7.68 • Silence/Noise +6.12 • Unknown +5.77
• Music +3.62

1-FRAME • MINIMAL VISUAL INFO

+0.56 pp

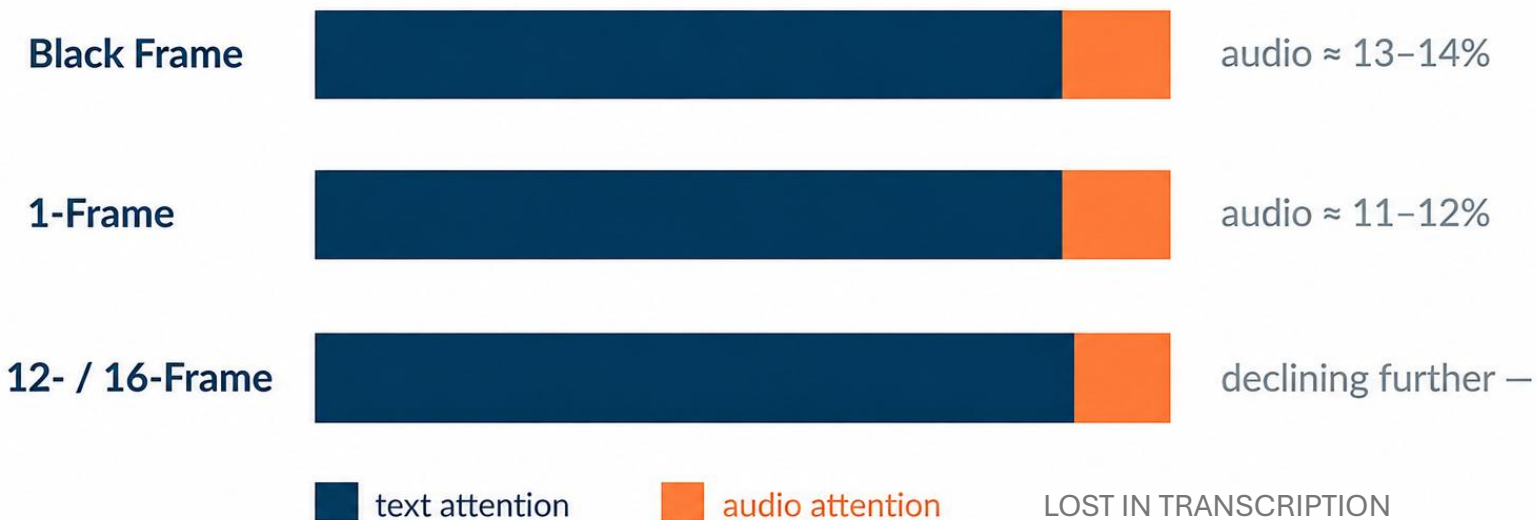
78.84% → 79.40%

only marginal overall • $p = .032$, CI crosses 0

NOT significant in any single category after Holm correction — e.g. Dialogue +0.45
($p_{\text{Holm}} = .714$).

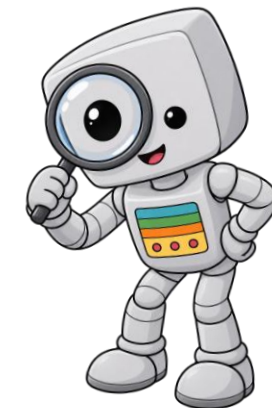
One static frame is already enough to make audio's contribution **mostly vanish**.

Attention Allocation & the Limits of Fusion



- Adding audio **reallocates 11–14% of attention away from text**, the model implicitly modulates audio's relevance to how much visual information is available
- **But more attention does not mean better reasoning:** in the 1-Frame setting, audio still gets 11–12% of the budget despite non-robust accuracy gain

Key Findings



**Visual Evidence
remains the
Dominant**



**Under dense Visual
Evidence:
raw audio $\approx$ no
effect**



**Transcriptions cause
a moderate but
systematic
interference**

Open question:

Is this a property of MAIA's visually-grounded design, or a general limitation of how current architectures gate modalities?

Powering down...

Thank you for your *human* attention

